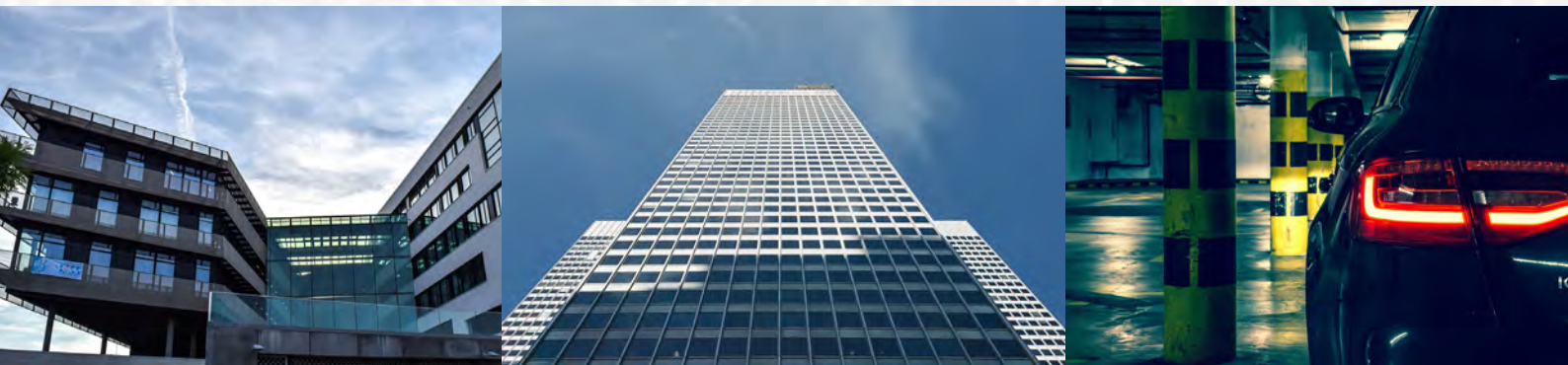




MODIFIED POLYISOCYANURATE RIGID THERMAL INSULATION
HIGH R-VALUE INSULATION PANEL



PIRmax HR-PANEL
TECHNICAL DATA



PRODUCT INTRODUCTION

PIRMAX HR PANEL is a modified PIR (Polyisocyanurate) rigid thermal panel that can be used in many applications. **PIRMAX HR PANEL** is the next generation in high performing insulation products for commercial and residential developments. Designed to achieve excellent thermal value where minimal cavity is available.

PIRMAX HR PANEL has amongst the highest thermal values of PIR insulation products, due to the low thermal conductivity, giving buildings continued thermal efficiency throughout their life span. **PIRMAX HR PANEL** is a light weight closed cell Polyisocyanurate foam core panel. Easy to handle and allows fast fixing to any application. Fully compliant to the Australian National Construction Code.



PIRMAX HR PANEL is CFC free with zero Ozone Depletion Potential (ODP) and contains zero Volatile Organic Compounds (VOC's) and is manufactured in Australia.

- ▶ Complies with AS/NZS 4859.1.
- ▶ Australian made & utilising chemicals and raw materials from Australian companies.
- ▶ Amongst the highest thermal values of rigid board insulation due to the product's low thermal conductivity.
- ▶ Achieves a Group 1 BCA/NCC fire classification when tested to AS ISO 9705:2003, compliant to the 2016 National Construction Code.
- ▶ Not corrosive to metal over time.
- ▶ Able to custom manufacture to different sizes and facing colours.
- ▶ Fast manufacturing turnaround times.

Applications

Soffit (Carpark Application), Walls External/Internal, Raised Underfloor, Ground Floor Slabs, Ceilings, Roofs, Domestic and Commercial Ducting, External Cladding.

Construction

PIRMAX HR PANEL is a modified PIR (Polyisocyanurate) rigid thermal panel laminated with aluminium foil laminates to both sides of the board. The foil laminates have a reflectivity of 94% and emissivity of 0.06.

Australian Made

PIRMAX HR PANELS are manufactured by an Australian owned and operated company, and guarantee the highest quality insulation material. We also source our raw materials and chemicals from Australian companies wherever possible. As our products are wholly Australian Made we take pride in our manufacturing to ensure a high standard of quality, service and value.



PRODUCT SPECIFICATIONS

PIRMAX HR PANEL Product Details

Product Name	PIRMAX - HR Panel "modified PIR rigid thermal insulation"
Product Code	PIRMAX HR _____ (mm)
Thickness	20mm/25mm/30mm/40mm/50mm/60mm/70mm/75mm/80mm/90mm/100mm
Panel Size	2400mm x 1200mm (2.88m ²) WA: 2270mm x 1200mm (2.72m ²); 514mm x 2270mm (1.166m ²) Also can be cut to custom lengths and sizes
Facings	Silver/White/Black. Embossed aluminium sheet laminated to both sides. (Other colours are available upon request with an MOQ required).

PIRMAX HR PANEL Product Data

Thermal Conductivity:	0.020 W/mK
Emissivity	E0.06 (silver foil face)

PIRMAX HR PANEL Product R-Values

Product Thickness:	Product R-Value
PIRMAX HR 20mm	R1.0
PIRMAX HR 25mm	R1.25
PIRMAX HR 30mm	R1.5
PIRMAX HR 40mm	R2.0
PIRMAX HR 50mm	R2.5
PIRMAX HR 60mm	R3.0
PIRMAX HR 70mm	R3.5
PIRMAX HR 75mm	R3.8
PIRMAX HR 80mm	R4.0
PIRMAX HR 90mm	R4.5
PIRMAX HR 100mm	R5.0

PIRMAX HR PANEL Product Testing

Complies to AS/NZ 4859.1	
NCC/BCA Compliant	
Group Classification AS ISO 9705:2003	Group 1 Lining
Group Classification AS/NZS 3837:1998	Group 1 Lining
AS 1530.3 Regulatory Indices:	Ignitability Index 0 Range 0-20
	Spread of Flame Index 0 Range 0-10
	Heat Evolved Index 0 Range 0-10
	Smoke Developed Index 1 Range 0-10
Compressive Strength	>125 (kPa)
Water Vapour Resistivity	>100 (MNs/gm)
Edge Treatment	Square Edge

PIRMAX HR PANEL Specification Notes

When specifying, state the following:

Product Name: PIRMAX HR PANEL _____mm

The insulation to be installed shall be PIRMAX HR Panel double sided reflective PIR insulation, comprised of Modified Polyisocyanurate Rigid Panel with foil facing on both sides, thermal resistance with an R Value of _____. Product is manufactured by PIRMAX and shall be installed in accordance with manufacturers installation instructions.



GENERAL INSTALLATION GUIDE

Under Soffit Installation

INSULATION FASTENERS

PIRMAX HR Panel can be fully secured to a concrete soffit by the use of approved insulation fasteners. 5 insulation fasteners can be used in suspended ceiling applications or where the boards will not be subject to external wind pressure or fire ratings, however 11 insulation fasteners is recommended and the insulation fasteners must have a minimum head diameter of 35mm.

The material of the fixings should also be appropriate for the application, exposure and fire rating as advised by the fixing manufacturer.

DIRECT FIXING TO SOFFITS

1. The fasteners should be evenly distributed over the whole area of the board. Install 4 fasteners evenly across each length, no less than 50mm and no greater than 150mm in from the edge.

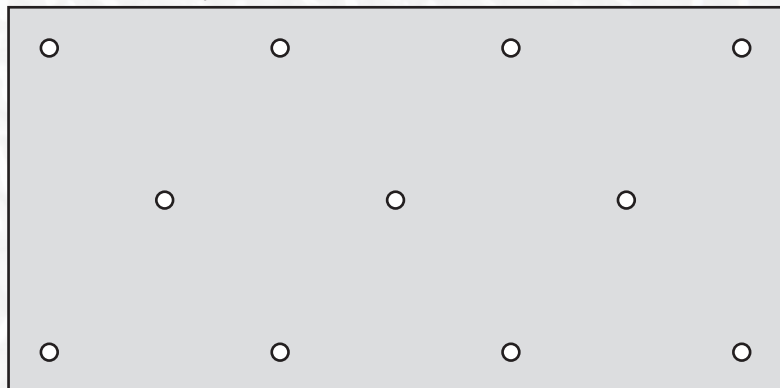
Install 3 fasteners along the centre of the board as per the diagram.

2. Board joints can be either staggered or squared and taped with a minimum 75 mm wide high quality aluminium foil tape. When taping, ensure the surfaces are clean, sound and free of dust and contaminants. Use a tape applicator to apply appropriate pressure to the tape.
3. Installers should check their specific project to make sure there are not any non-standard fixing requirements. Where the board may be subject to external wind pressure, the requirement for additional fixings should be assessed in accordance with appropriate Australian wind load standards.

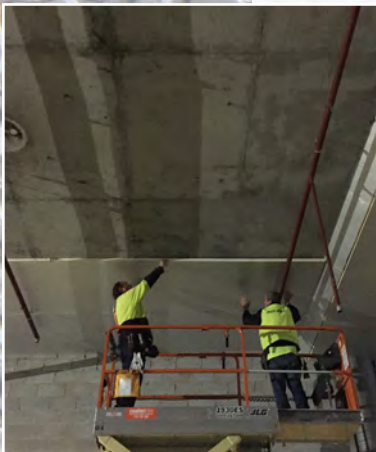
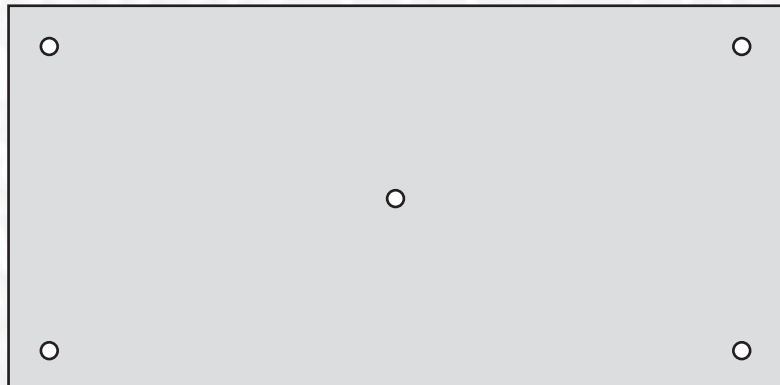
PENETRATIONS

If there are fire collar penetrations in the soffit, then a clearance of a minimum of 25mm must be maintained between the fire collar and the insulation board.

11 insulation fastener pattern (recommended)



5 insulation fastener pattern (suspended ceiling/no external wind pressure applications)



PIR PANELS
www.pirmax.com.au

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